

		$\lambda \propto \frac{1}{\sqrt{m}}$ or $\frac{p}{\lambda_e} = \sqrt{\frac{m_e}{m_p}} = \sqrt{\frac{9.11 \times 10^{-31}}{1.67 \times 10^{-27}}} = 0.0234$
29.	C	Original count rate per minute due to source = $532 - 24 = 508$ per minute After two half-lives, the count rate would have dropped to $0.5 \times 0.5 \times 508 = 127$. Taking the background count rate into consideration, the reading will be $127 + 24 = 151$.
30.	D	Since the radiation is able to pass through 1 mm of aluminium sheet, it cannot